

Polarimeter

1 Aim

1. To study the optical rotation caused by chiral molecules using a polarimeter setup and to find the specific rotation of a solution of sugar using the polarimeter.
2. To find the unknown concentration of a solution from its optical activity.

2 Introduction

Light waves associated with light produced from a source such as a candle or a light bulb are unpolarized i.e. the electric field oscillates in random directions orthogonal to the direction of propagation. A polarization filter is a device with a specified direction called the polarization axis. The filter absorbs all the incoming waves electric field component perpendicular to the polarization axis and transmits the electric field parallel to the polarization axis (linearly polarized light). An unpolarized wave just before passing through the polarizer has its electric field pointing randomly in all directions perpendicular to the direction of wave propagation. After passing through the polarizer, the wave has only half the energy, with the electric field oscillating solely along the polarization axis. The most general form of polarized light is elliptically polarized light, with circularly polarized and linearly polarized light being special cases.

Certain optically active substances such as quartz, turpentine and sugar solution can rotate the plane of polarization of a plane polarized light wave passing through the substance. Dextro-rotatory (right-handed) substances rotate the plane of polarization toward the right (in the clockwise direction on looking toward the source) while laevo-rotatory (left-handed) substances rotate the plane of polarization toward the left (in the anti-clockwise direction on looking toward the source). The angle by which the plane of polarization is rotated depends upon the concentration of the solution (or density of substance), the wavelength of light and the temperature. The specific rotation S at a given temperature and wavelength of light is defined as the optical rotation produced

by one decimetre length of solution of unit concentration.

$$S = \frac{\theta}{lc},$$

where l is the length of the polarimeter tube in decimetre, m is the mass of the material dissolved in water and c is the concentration in gm/cc.

A nicol prism is an optical device made from calcite and can be used to produce and analyze plane polarized light. When light is passed through a doubly refracting crystal it splits into an ordinary and an extraordinary part, both plane polarized. In the nicol prism, one of these rays is cut off by total internal reflection.

Laurents half-shade polarimeter is an instrument used to find the optical rotation of optically active solutions. The light from a source (sodium vapor lamp) passes through a narrow slit and is made parallel by a lens. The light is plane polarized by a nicol prism (called the polarizer) and then passes through a half-shade device and then a glass tube containing the optically active solution. The light then falls on another nicol prism (called the analyzer) and the emergent light is viewed through a telescope. The analyzer can be rotated about the tube axis and its rotation measured on a graduated circular scale. The position of the analyzer is adjusted such that the field of view is completely dark. The tube is refilled with the optically active solution and inserted back in position. The field of view now becomes illuminated and darkness can be achieved by rotating the analyzing nicol prism through a certain angle depending on the optical rotation of the solution.

3 Apparatus in the lab and procedure

1. Clean the polarimeter tube and place the polarimeter in front of a sodium vapor lamp. Look through the telescope and adjust the eyepiece so that the two halves of the half-shade circle are clearly in focus.
2. Fill the glass tube with distilled water, position it inside the polarimeter and cover it. Rotate the analyzer by rotating the circular scale till the two halves of the half-shade device are equally dark. In this position, there will be a drastic change in intensity of the two halves for a slight rotation on either side. Note the reading on the scale.
3. Rotate the analyzer by approximately 180° and record the other equally dark position reading. Keep rotating by 180° and note down all the other readings for equally dark halves.
4. Find out the room temperature.
5. Find the least count of the analyzer (as the value of one division of the main circular scale in degrees divided by the total number of divisions of the vernier scale).



Figure 1: Apparatus in the Lab.

6. Fill the tube with sugar solution of a known concentration and again note down the scale readings when the two halves are in an equally dark position. Repeat for different concentrations of the sugar solution.
7. Plot a graph between θ and the concentration C . Do a least squares fit and obtain the slope and calculate the specific rotation at room temperature.
8. Repeat the experiment for a solution with unknown concentration from the graph plotted between θ and C , obtain the unknown concentration.

4 Precautions

1. The glass tube should be cleaned.
2. There should be no air bubble in the glass tube when filled with solution. Before filling

4.1 Points to Ponder

1. Can you use fruit juice for this experiment?